Cambridge Assessment International Education

Cambridge IGCSE™

CANDIDATE NAME	

CENTRE NUMBER		CANDIDATE NUMBER	

PHYSICS **Pre-IG**

Pre-IG Theory **June 2026**

You must answer on the question paper.

No additional materials are needed.

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- You should show all your working and use appropriate units.
- Take the weight of 1.0 kg to be 10 N (acceleration of free fall = 10 m/s^2).

INFORMATION

- The total mark for this paper is 40.
- The number of marks for each question or part question is shown in brackets [ ].

This document has 14 pages. Any blank pages are indicated.

1 A digital stop-clock measures time in minutes and seconds.

The stop-clock reads 00:50 when it is started (i.e. 00 minutes 50 seconds).

It reads 02:10 when it is stopped.

What is the shortest possible time that has elapsed between starting and stopping the stop-clock?

A 1 minute 20 seconds

B 2 minutes 00 seconds

C 2 minutes 10 seconds

D 3 minutes 00 seconds

2 A long-distance runner wishes to calculate her average speed for a race.

Which calculation should she use?

A average speed $= \dfrac{\text{total distance}}{\text{total time}}$

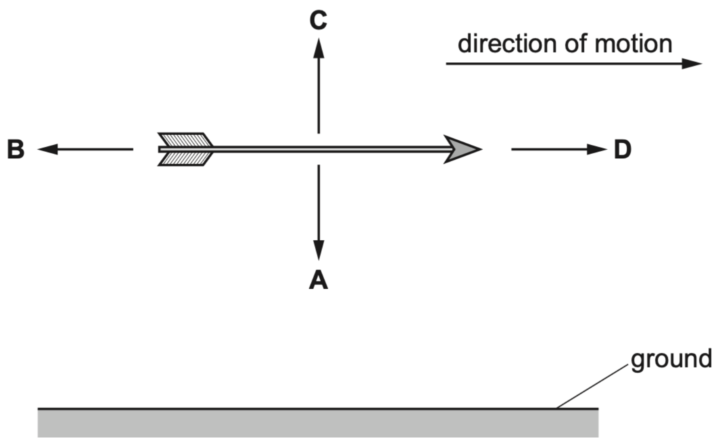
B average speed = total distance $\times$ total time

C average speed $= \dfrac{\text{total time}}{\text{total distance}}$

D average speed = total distance + total time

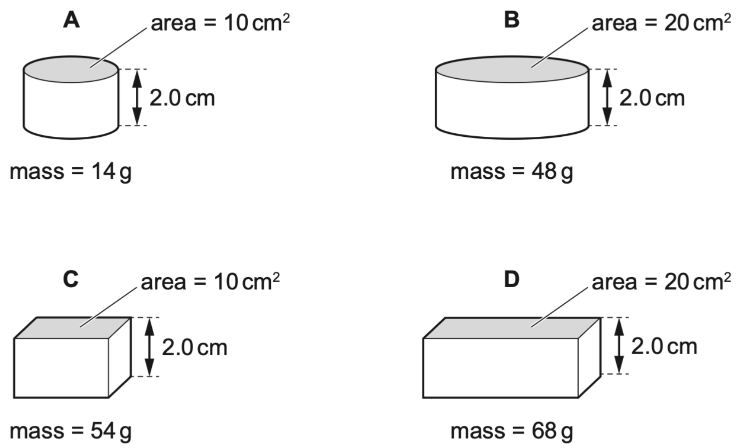
3 An arrow travels horizontally in a straight line at constant speed.

In which direction does the weight act?



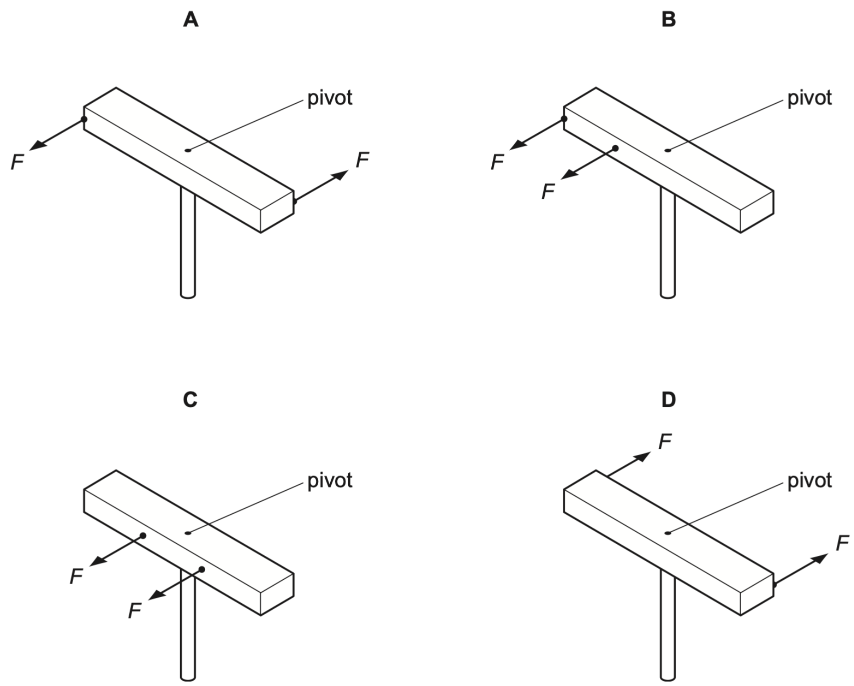
4 The diagrams show four solid blocks with their dimensions and masses.

Which block has the greatest density?

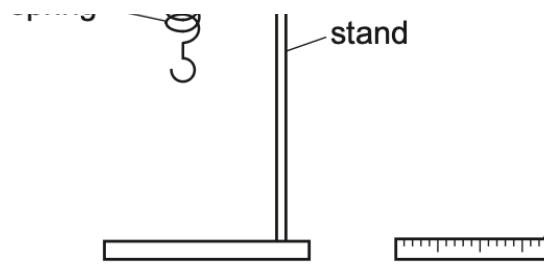


5 A wooden bar is pivoted at its centre so that it can rotate freely. Two equal forces F are applied to the bar.

In which diagram is the turning effect greatest?



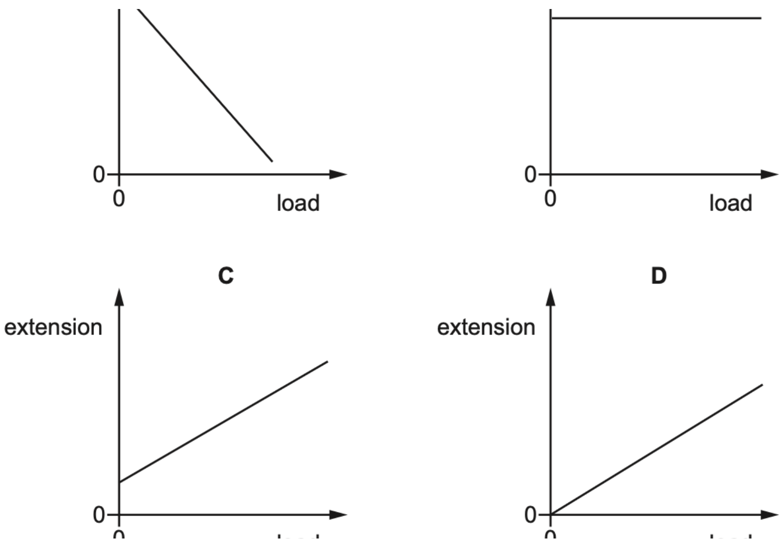
6 A spring is suspended from a stand. Loads are added and the extensions are measured.



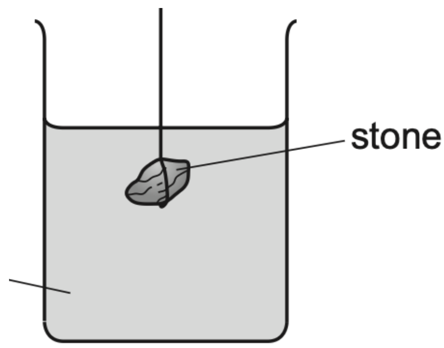
lt of plotting extension against load?

A

Which graph shows the result of plotting extension against load?



7 The diagram shows a stone suspended on a string under the surface of a liquid. The stone experiences a pressure caused by the liquid.



What would increase the pressure on the stone?

A decreasing the surface area of the stone

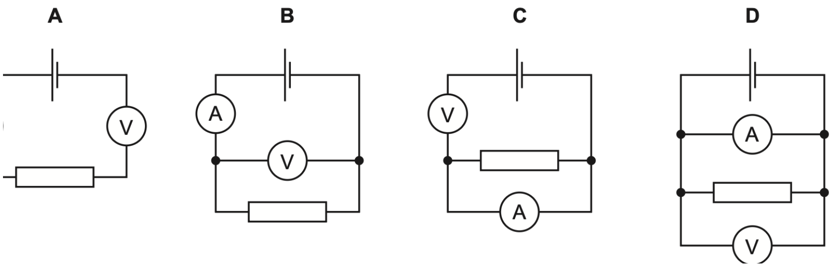
B increasing the mass of the stone

C lowering the stone deeper into the liquid

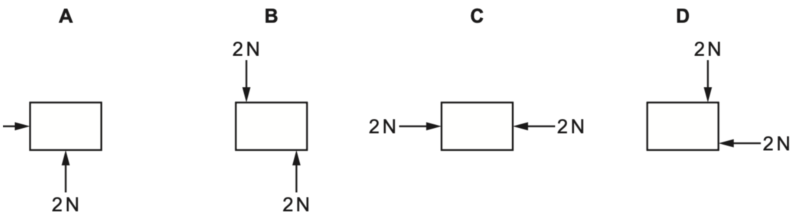
D using a liquid with a lower density

8 A voltmeter and an ammeter are used to measure the resistance of a resistor.

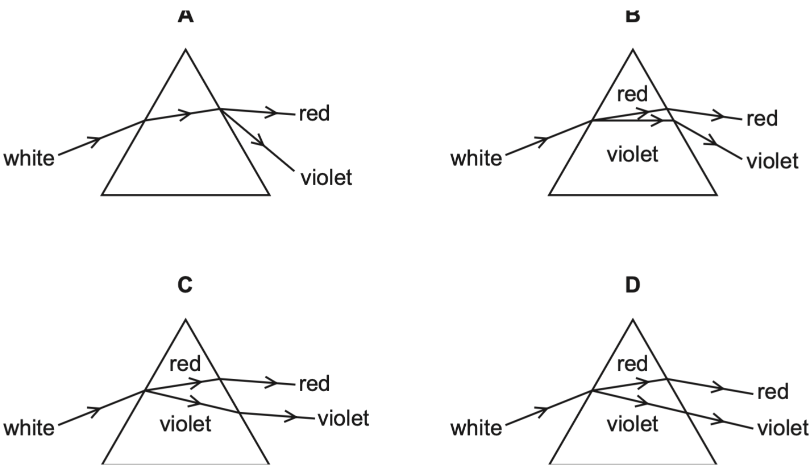
Which diagram shows the voltmeter and the ammeter correctly connected?



9 Which object is in equilibrium?

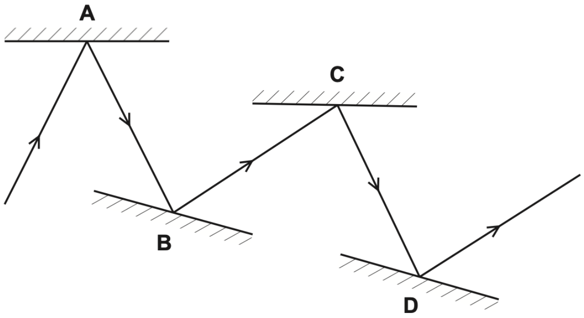


10 Which diagram shows the dispersion of white light by a glass prism?

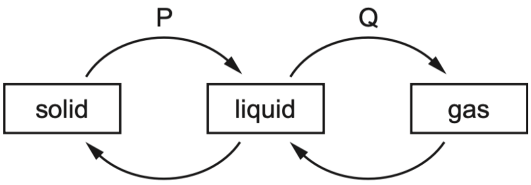


11 A student draws a ray diagram to show how a ray of light is reflected by a number of mirrors.

Which reflection has not been drawn correctly?



12 The diagram shows four labelled changes of state between solid, liquid and gas.



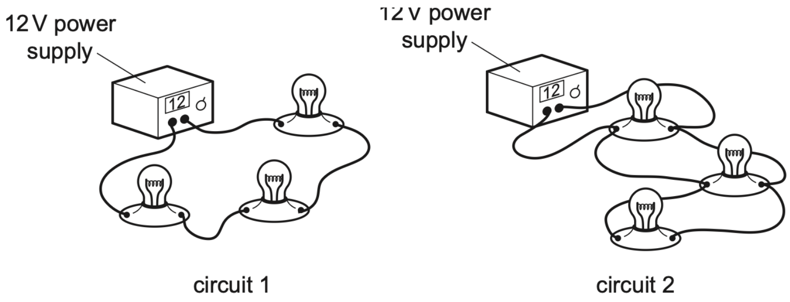
Which changes need an energy input?

A P and Q **B** Q and R **C** R and S **D** S and P

13 A student is designing a lighting circuit for a dolls' house. He sets up two different circuits.

Each circuit contains a 12V power supply and three identical lamps.

Each lamp is designed to operate at normal brightness when connected individually to a 12V supply.



Which statement is correct?

A In circuit 1, each of the lamps is at normal brightness.

B In circuit 1, if one lamp fails, the other lamps remain lit.

C In circuit 2, if one lamp fails, the other lamps remain lit.

D In circuit 2, the current from the power supply is less than in circuit 1.

14 **(a)** The arrows on Fig. 14.1 represent changes of state.

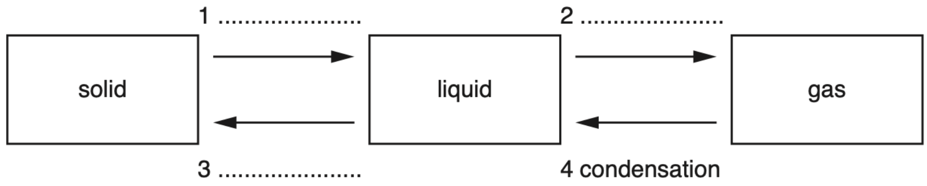


Fig. 14.1

One of the arrows is labelled. Label each of the other arrows with the correct change of state.

Write the change of state on the dotted lines next to each arrow.

[3]

[Total: 3]

15 **(a)** A student has a metal object.

(a) **(i)** The student measures the mass of the object.

State the name of the equipment used to measure the mass.

.... [1]

(ii) The mass of the metal object is 1260g. The volume of the metal is $150cm^3$.

Calculate the density of the metal. Include the unit.

density = ... [4]

(iii) The mass of the metal object is given in grams. State the mass in kg.

mass = ... kg [1]

(b) A vase is placed on a table. Forces X and Y act on the vase, as shown in Fig. 15.1.

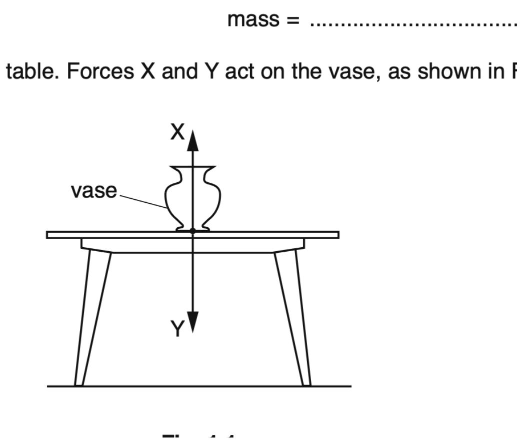


Fig. 15.1

The mass of the vase is 0.25kg. The vase is not moving.

Calculate the value of force X and the value of force Y.

X ....

Y ....

[4]

[Total: 10]

16 **(a)** A student rubs a plastic rod with a dry cloth, as shown in Fig. 16.1. The rod becomes negatively charged.

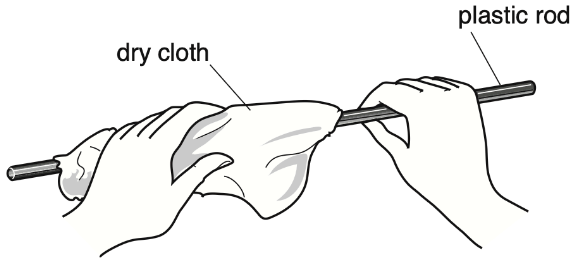


Fig. 8.1

Fig. 16.1

(a) **(i)** Use words from the box to complete the sentence.

air	cloth	electrons	hand	neutrons	protons

The rod becomes negatively charged because .... move from the to the rod. .... [2]

(ii) The student moves the rod close to a suspended, charged rod. The two rods repel each other.

State the type of charge on the suspended rod.

.... [1]

(iii) Explain your answer to (a)(ii).

....

.... [1]

(b) A device has a metal case. Any charge on the case must be able to move to earth.

(b) **(i)** Draw one ring around a material that is suitable for the connection to earth.

copper glass plastic rubber

[1]

(ii) Explain your answer to (b)(i).

....

.... [1]

[Total: 6]

17 Fig. 17.1 shows a ray of light that is reflected by a mirror.

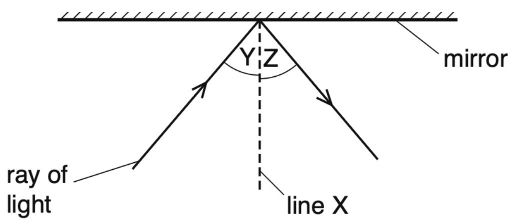


Fig. 17.1

(a) **(i)** State the name of line X shown on Fig. 17.1.

.... [1]

(ii) State the name of angle Y shown on Fig. 17.1.

.... [1]

(iii) A student moves the ray of light and doubles the size of angle Y. State the effect on angle Z.

.... [1]

(b) Fig. 17.2 shows a converging lens used to form an image I of an object O.

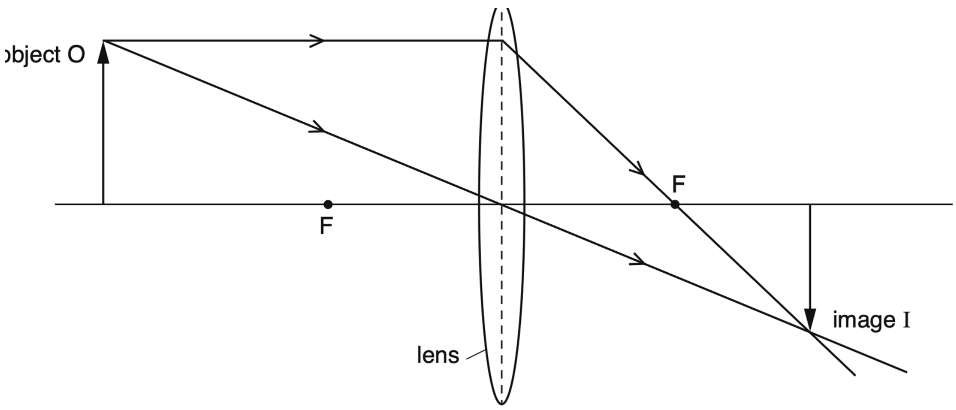


Fig. 6.2

Fig. 17.2

(b) **(i)** State the name of the points labelled F on Fig. 17.2.

.... [1]

(ii) Describe the nature of the image I.

....

....

.... [2]

[Total: 6]

18 (a) State the two conditions which must be true for an object to be in equilibrium.

condition 1 ....

condition 2 .... [2]

(b) Fig. 18.1 shows a uniform metre rule PQ in equilibrium.

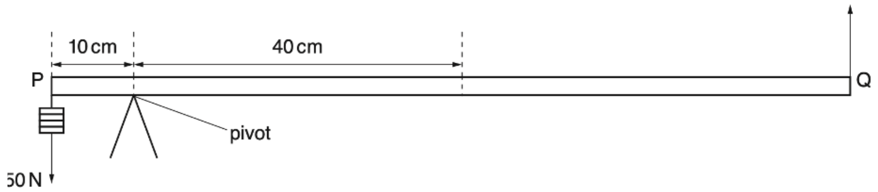


Fig. 18.1

The distance PQ is 100cm. The mass of the metre rule is 0.12kg and its weight is W.

(b) (i) On Fig. 18.1, draw and label:

1. an arrow to show the force W acting on PQ at the centre of mass
2. an arrow to show the force R acting on PQ at the pivot.

[2]

(ii) By taking moments about the pivot, calculate F.

F = ... [4]

(iii) Calculate R.

R = ... [2]

[Total: 10]